



Master Thesis

Thesis Title: Concise and Engaging Title

by

Wouter Bakker
(2655935)

First Supervisor: Zhao Yang

Second Reader: Name and Surname

June 4, 2025

Submitted in partial fulfillment of the requirements for
the VU degree of Master of Science in Artificial Intelligence

LLM Evaluators: teststs

Wouter Bakker

Vrije Universiteit Amsterdam, Amsterdam `w4.bakker@student.vu.nl`

Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

1.1 General Introduction

DRAFT

LLMs are increasingly used for content generation. However, reliability remains a bottleneck.

Subjective evaluation poses a challenge for Large Language Models (LLMs) and humans alike. Consider an evaluation task, such as the correctness of a mathematical derivation, the number of grammatical errors in a sentence, or the helpfulness of an LLM output. An evaluation task is subjective when the ground truth is not explicitly known, such as for the helpfulness of LLM output: what is helpful to one user might be irrelevant to another. Contrast this with objective evaluation, where output can be validated against an observable ground truth, as is generally the case for e.g. mathematical statements and grammar.

General practice in subjective evaluation is to ask humans to provide 'gold labels'. By training humans to follow a customized evaluation protocol, human evaluators are assumed to approximate a valid measurement of the latent ground truth. However, this process is unscalable.

To automate evaluation, classic Natural Language Generation (NLG) evaluation techniques such as BLEU and ROUGE were often used. Yet, because these only consider surface-level linguistic characteristics like n-gram overlap, they often suffer from low correlation with human evaluations [5]. While useful for simple tasks like translation and summarization quality, they fail to capture the linguistic nuances needed for more complex tasks.

LLMs offer a promising solution. In a seminal paper, Zheng et al. [12] use well-aligned LLMs such as GPT4 in pairwise comparisons, where two candidate texts are compared and the best one is selected. GPT-4 achieves over 80% agreement with human preferences, on par with human-human agreement. Liu et al. [5] adopt a similar approach in their G-Eval framework, prompting LLMs to grade text on a fine-grained Likert scale. Their approach shows significantly higher correlation with human judgements compared to previous state-of-the-art evaluators.

However promising, (limitations of current methods, bridge to subsection 1.2 industrial partner operational challenges)

1.2 Industrial Partner: Barter

The current study is motivated by the operational challenges faced by the industrial partner, Barter. The company aims to leverage Generative AI (GenAI) to streamline the onboarding experience, specifically addressing the friction new partners experience during the sign-up flow. Currently, users must manually write marketing copy such as company descriptions and deal offers before they can make use of the platform. This creates a significant barrier to entry, resulting in user attrition.

While GenAI offers a scalable solution to automate this drafting process, its implementation in business contexts is hindered by the inherent variability of output quality. As LLMs are probabilistic in nature, some degree of inconsistency is unavoidable. Although well-engineered prompts might yield high-quality output in the majority of cases, the accumulation of low-quality output (such as hallucinations and misalignment) becomes a significant operational problem at high volumes.

Two fundamental challenges lie at the core of this problem. First is the challenge of identification: without a reliable mechanism to flag low-quality output, we are unable to quantify the severity of the problem, or make the problematic output actionable. The second is the challenge of diagnosis: improving the output requires an evaluation to identify the reasons why the quality is insufficient. For GenAI to function as an automated solution, it must not only detect the errors but also provide the feedback to resolve them.

Therefore, although Barter's problem statement concerns generation, the solution to the problem begins with accurate validation. Currently, Barter operates in a similar fashion, where employees use GenAI to generate content which is manually evaluated and refined. However, such a human-in-the-loop (HITL) approach is costly and unscalable. To address this, this thesis explores whether small-scale open-source LLMs can be used for automated "AI-in-the-loop" (AITL) evaluation. We introduce a method inspired by the latent variable models of Psychometrics to provide the granular evaluation required for the identification and diagnosis of output quality.

2 Literature Review

2.1 Traditional NLP evaluation methods

2.2 LLM-as-a-Judge

2.3 Psychometrics in NLP

3 Theoretical Framework

3.1 Measurement view of LLM-as-a-Judge

This thesis treats LLM-based evaluators as measurement instruments that map an input (e.g., marketing copy text) to a score that represents an underlying (subjective) construct (e.g., "Quality"). To formalize this, we adopt the framework of Psychometric Latent Variable Theory. Psychometrics concerns itself with the measurement of psychological attributes, called constructs. In psychology, constructs are defined as latent (unobservable) variables underlying human behavior, such as cognitive abilities and personality traits. Assuming constructs can be treated as quantitative (for a discussion, see: [7]), we can measure them by defining their relationship with manifest (observable) variables ("indicators"). Typically, these manifest variables are measured through item scores on a test. For example, the latent construct of *Extraversion* is measured via items like "Talks a lot" or "Has a lot of energy" [3].

The same theoretical principles underlie measurements that are used in Machine Learning (ML), where "gold" labels for supervised learning are sourced by having people rate samples. By supplying the raters with specific criteria, their rating is assumed to be a reliable and valid representation of some underlying construct.

We hypothesize that a similar approach can be applied to LLMs. As Zheng et al. [12] and related LLMAJ literature demonstrate, LLMs are capable of mimicking human judgment in evaluation tasks. In psychometrical terms, this implies they are capable of measuring latent variables.

However, a latent variable is only meaningful relative to a rigorous construct definition. Adopting a latent variable framework allows us to methodically design an LLM evaluator and test its validity. Accordingly, this chapter is organized as follows:

- **Theoretical Definition (Section 3.2):** We explain the two causal models that define the relationship between the latent variable and its indicators: the *reflective* and the *formative* model.
- **Computational Operationalization (Section 3.3):** We demonstrate how a subjective evaluation task can be designed to measure these indicators, and how their measurement can be operationalized by using LLMs as instruments.
- **Empirical Validation (Section 3.4):** We use the psychometrical concept of *construct validity* and its subtypes to establish a framework for empirically validating the instrument.

3.2 Latent Variables Models: Reflective vs. Formative

Latent variable models define a relationship between manifest and latent variables. Two model types are distinguished: formative and reflective [1]. They differ in their assumptions about the direction of causality.

In formative models, the manifest variables are assumed to be the common cause of the latent variable. The canonical example is Socioeconomic Status (SES), which can be defined as the composite of three main features: income, education, and prestige [6]. Formally, for a set of manifest variables indexed by $i = 1, 2, \dots, n$, we assume

$$\eta = \sum_{i=1}^n \gamma_i x_i + \zeta \quad (1)$$

where η is a latent construct, γ_i a formative weight, x_i a manifest variable, ζ the disturbance term.

In reflective models, the latent variable causes the manifest variables. For example, following a "g-factor" interpretation of the latent variable intelligence [9], we assume that intelligence is an inherent trait that humans possess which causes people's performance on cognitive tasks. Formally, for any given manifest variable i , we assume

$$x_i = \lambda_i \eta + \varepsilon_i \quad (2)$$

where x_i is a manifest variable, λ_i a factor loading, η the latent construct, and ε_i the measurement error.

These definitions have consequences for construct operationalization. In the formative model, the latent variable is defined by its constituents. Therefore, omitting one variable changes the meaning of the latent variable (e.g., measuring "SES" without "Income" fundamentally changes the construct). Conversely, in the reflective model, the constituents are the effect of the latent variable, and are therefore assumed to covary. Omission of one of its constituents would not redefine the construct.

3.3 Operationalizing LLM Evaluation

Assuming the construct to be formative or reflective determines how to design the measurement instrument and how to test for validity. Construct misspecification is a highly debated topic in psychology and the social sciences. Typically, stemming from the field of Classical Test Theory, constructs were implicitly assumed to be reflective [8]. As the effect indicators are assumed to be caused by the latent construct,

We consider the "Quality" construct ("Q&A answer quality" in the case of Li et al. [4], and "marketing copy quality" in the case of Barter Deals).

Measurement instrument validity depends on whether the latent construct is assumed to be formative or reflective. In the formative model, the latent variable is defined by the indicators, so "omitting an indicator is omitting a part of the construct" [1]. Omitting one indicator changes the meaning of the latent variable (e.g., measuring "SES" without "Income" fundamentally changes the construct). Conversely, in the reflective model, the indicators are the effect of the latent variable. This implies that the indicators must covary. Omitting one indicator would not redefine the construct, and they are interchangeable [2].

Consider the "Quality" construct ("Q&A answer quality" in the case of Li et al. [4], and "marketing copy quality" in the case of Barter Deals). To determine whether *Quality* is formative or reflective, we consider the direction of causality. Consider some indicators of text quality, such as fluency, tone, and relevancy. If *Quality* were reflective, an increase in *Quality* would cause an increase in all its indicators. However, this is theoretically incoherent. For example, by improving text relevancy *Quality* can be improved, while retaining the same level of fluency. On the other hand, if we improve fluency, *Quality* increases. Therefore, *Quality* is considered to be a formative construct.

Explain how an evaluation task can be framed as a formative latent variable measurement problem, and how we use LLMs as measurement devices.

Explain the processing pipeline: we define the formative variable, and measure the subdimensions individually. We design a prompt that targets these subdimensions specifically. We have the LLM process the prompt (here there is a tree: some use CoT, reasoning models, etc., but we opt for single-token generation: it functions as a 'snapshot' of subjectivity, and is computationally more efficient, which is a very strict requirement if a model is to be used in production; arguably this is more methods however). Internally, the LLM processes the prompt. Generation in LLMs works by computing the conditional probability of each token in the vocabulary, given the previous tokens. This produces logits, or scores, which can be converted to probabilities using the softmax function. A distinction should be made between single-token generation and multi-token generation. Given a prompt, the logits are computed for the first next token. When the temperature $T = 0$, the token with the highest logit in the set of allowed tokens is selected. This set can be the full token set (the entire vocabulary) for unconstrained generation, or a subset for constrained generation. For evaluation, generation would be constrained to e.g. set(1,2,3,4), corresponding to the possible values on a four-point likert scale. Multi-token generation adds a layer of complexity. Although the logits are deterministically computed, sampling is generally done stochastically. For example, first selecting low logit tokens might lead to a higher marginal probability if it leads to a sequence of high logit tokens. Methods like beam search allow efficient mapping of the probability space. However, for the current project, we specifically aim to investigate the logits, which are used to gain a better understanding of how the model processes the input. For multi-token generation, marginal probabilities would have to be estimated to obtain a reliable estimate of the model's internal representation of the construct, which adds a huge layer of complexity which we avoid. Moreover, single-token generation only requires a single forward pass, which is computationally cheap, and therefore best suited for production environments.

After a single forward pass, the model's internal representation reflects its subjective evaluation of the target text, contained in the subset of constrained tokens corresponding to the target evaluation space.

3.4 Construct Validity & Validation of the Evaluation Instrument

"Quality" is a latent variable == construct. Explain the broader concept of validity (by comparing with reliability, example of darts: if a robot throws 10 darts in exactly the same place but they're not on the dartboard, it is reliable, but invalid- confidently incorrect). Explain construct validity, "whether a test (or measurement device) measures what we intend for it to measure", discuss different subsumers that are relevant: 1. content validity (important when defining the subdimensions, and afterwards the prompts) 2. Criterion validity (in-depth explanation of how it relates its operationalization: FeedbackQA comparison with ground truth (concurrent validity), Barter prediction of KPI (predictive validity))

"Validity becomes actionable: each validity type is introduced at the moment it matters, tied to the concrete test (FeedbackQA, KPI prediction)."

- content validity = rubric coverage + prompt constraints - concurrent validity = agreement with human ratings on FeedbackQA - predictive validity = correlation/utility vs downstream KPI

"Ultimately, the validity of a measurement instrument cannot be established through theoretical design alone; it requires empirical verification against external criteria. Following the framework of Criterion Validity, we test the proposed formative evaluator on two distinct axes. First, we assess Concurrent Validity by measuring the alignment between the model's scores and human-annotated 'gold labels' within the FeedbackQA dataset. Second, we evaluate Predictive Validity by examining the relationship between the model's quality scores and downstream performance metrics (KPIs) within Barter's industrial environment. By satisfying both academic and industrial validation criteria, we aim to demonstrate that the formative latent variable approach provides a robust, scalable alternative to traditional holistic human oversight. The following chapter details the specific experimental setup, datasets, and model architectures used to perform this validation."

diagram of how an LLM evaluator assuming a formative model would work

4 Methods

4.1 Data

FeedbackQA The *FeedbackQA* dataset Li et al. [4] contains evaluations of Q&A pairs ($n = 5660$). Q&A pairs related to Covid-19 were scraped from the websites of WHO, US Government, UK Government, Canadian government, and the Australian government. Crowd-source workers from English-speaking countries were hired through Amazon’s MTurk platform to rate each Q&A pair and provide feedback. The rating is performed on a 4-point likert scale (*bad, could be improved, good, excellent*). The feedback concerns "a natural language explanation elaborating on the strengths and/or weaknesses of the answer" [4]. Since each Q&A pair contains two ratings, we aggregate both into the mean to obtain our "ground truth" target variable.

Barter deals The *Barter Deals* dataset contains 'deals' written by companies ($n = ???$). On the Barter platform, companies ('partners') can barter goods and services in exchange for influencer content. For example, a company that sells hair styling tools offers their new product in exchange for a video of a creator using it; or, a museum provides free entrance to a creator and their friends in exchange for a video of their visit. The 'deal text' is the marketing copy that is posted on the Barter platform. Users can apply to deals, and the partner can accept or reject the application. Once accepted, the partner provides the good or service, and the creator provides the content.

In the current study, we consider the 'deal text' (marketing copy) and the number of applications the deal has obtained after the first seven days of going live (*applications after 7 days*). The variable *applications after 7 days* is chosen because it serves as a proxy for marketing copy quality. One might apply to a deal because they like the product or the brand, but they could also be convinced to apply because of the way the copy is written. A valid measurement of marketing copy quality would therefore be predictive of whether somebody applies to a deal or not. By controlling for the fixed effects of the brand and the product category we can separate the effects of marketing copy quality, and hereby determine whether the LLM provides a valid measurement of it.

4.2 Operationalization of Evaluation Constructs

To empirically measure the efficacy of the formative prompting framework, we define an experimental condition and a control condition (baseline). In the experimental condition (*formative prompt*) we treat the construct as a formative variable by decomposing it into sub-dimensions (e.g., d_1, d_2, d_3). We prompt the LLM to measure the subdimensions individually, yielding multiple scalar values. In the control condition (*holistic prompt*) we treat the construct as a monolith, forcing the model to implicitly aggregate the dimensions into a single scalar rating Y .

Construct definition: FeedbackQA For the question answering task, we derive the formative dimensions from the qualitative analysis of the feedback provided in Li et al. [4]. The authors identify numerous criteria that define "answer quality" by examining the explanations crowd-sourced workers provided for their ratings. From this, we derive three dimensions:

- **Relevance:** The extent to which the answer avoids irrelevant content
- **Completeness:** Whether the answer covers all aspects of the query.
- **Directness:** Whether the answer directly answers the question.

We expect a degree of covariance among the dimensions, since high-quality answers generally score high across all three. However, the formative framework allows for orthogonal variation across the dimensions as well. For example, Li et al. [4] highlight an answer rated *could be improved* (2/4) with the feedback: "*The answer is about some of the online risks but not about how to protect against them.*". This indicates an answer that is relevant but incomplete. While a holistic prompt forces the model to compress the conflicting signals into a single scalar value, the formative prompt explicitly models this by decomposing it.

Construct definition: Barter deals DRAFT:

In the absence of prior academic literature for the proprietary Barter dataset, we derived the dimensions through stakeholder analysis and domain requirements. The "Marketing Effectiveness" construct is decomposed into:

- **Persuasiveness:** The strength of the call-to-action.
- **Tone Alignment:** The consistency with the brand's requested voice.
- **Clarity:** The readability of the deal proposition.

4.3 Model selection

To perform evaluations we select three popular small-scale, open-source instruction-tuned models: Qwen3-4B-Instruct-2507 [11] (AliBaba), Gemma3-4B-IT [10] (Google), and Llama-3.2-3B-Instruct (Meta). These models were selected to represent the current state-of-the-art open-source models.

The choice of model type (small, open-source, instruction-tuned) is based on the methodological requirements, and the operational constraints of the industrial partner, Barter. Methodologically, the use of open-source models enables "white-box" access to the model's probability distribution. This allows us to compute the logit-weighted mean and the Shannon entropy. Moreover, we prefer instruction-tuned models over base models to ensure adherence to the evaluation instructions provided in the prompt. Operationally, Barter requires the evaluator to balance validity with cost-efficiency and latency. Proprietary models like GPT-4 incur costs that scale with volume, while open-source models incur only fixed compute costs, making them a more scalable solution.

4.4 Experimental design & Validation Strategy

Internal Validity: Alignment with Human Raters (FeedbackQA)

External Validity: Behavioral Prediction (Barter Deals)

5 Results

5.1 Academic validation (FeedbackQA)

5.2 Industrial application (Barter)

6 Discussion

"We assumed robustness; future work should stress-test it."

"A limitation of this study is the absence of adversarial robustness testing. While the method demonstrates high concurrent validity on the FeedbackQA dataset, we did not evaluate the model's stability against adversarial perturbations (e.g., randomized word order or irrelevant insertions). Future work should investigate whether the formative latent variables remain stable under such conditions to rule out spurious correlations."

Bibliography

- [1] K. Bollen and R. Lennox. Conventional wisdom on measurement: A structural equation perspective. *Psychological bulletin*, 110(2):305, 1991.
- [2] C. B. Jarvis, S. B. MacKenzie, and P. M. Podsakoff. A critical review of construct indicators and measurement model misspecification in marketing and consumer research. *Journal of consumer research*, 30(2):199–218, 2003.
- [3] O. P. John, E. M. Donahue, and R. L. Kentle. Big five inventory. *Journal of personality and social psychology*, 1991.
- [4] Z. Li, P. Sharma, X. H. Lu, J. C. K. Cheung, and S. Reddy. Using interactive feedback to improve the accuracy and explainability of question answering systems post-deployment. In *Findings of the Association for Computational Linguistics: ACL 2022*, pages 926–937. <https://doi.org/10.18653/v1/2022.findings-acl.75>. URL <http://arxiv.org/abs/2204.03025>.
- [5] Y. Liu, D. Iter, Y. Xu, S. Wang, R. Xu, and C. Zhu. G-eval: NLG evaluation using GPT-4 with better human alignment. URL <http://arxiv.org/abs/2303.16634>.
- [6] S. McQuitty and M. Wolf. Structural equation modeling: A practical introduction. *Journal of African Business*, 14(1):58–69, 2013.
- [7] J. Michell. Quantitative science and the definition of measurement in psychology. *British journal of Psychology*, 88(3):355–383, 1997.
- [8] C. Nunnally James and H. Bernstein Ira. Psychometric theory, 1996.
- [9] C. Spearman. "general intelligence" objectively determined and measured. 1961.
- [10] G. Team. Gemma 3. 2025. URL <https://goo.gle/Gemma3Report>.
- [11] Q. Team. Qwen3 technical report, 2025. URL <https://arxiv.org/abs/2505.09388>.
- [12] L. Zheng, W.-L. Chiang, Y. Sheng, S. Zhuang, Z. Wu, Y. Zhuang, Z. Lin, Z. Li, D. Li, E. Xing, H. Zhang, J. E. Gonzalez, and I. Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. URL <https://www.semanticscholar.org/paper/Judging-LLM-as-a-judge-with-MT-Bench-and-Chatbot-Zheng-Chiang/a0a79dad89857a96f8f71b14238e5237cbfc4787>.